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Paleomagnetism of Permian through Early Triassic sequences in central Spitsbergen: implications for paleogeography

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Abstract

Upper Carboniferous through Lower Triassic sediments from Spitsbergen have yielded paleomagnetic results, with those for the Late Gzhelian–Early Permian being the more reliable. The corresponding pole (38ºN, 171ºE) is fully concordant with the Early Permian segment of the apparent polar wander path for Baltica. It does not support hypotheses about post-Carboniferous mobilism of Spitsbergen with respect to Stable Europe. Neither Mesozoic nor later rifting processes changed the position of Spitsbergen considerably in relation to the Stable Europe. © 1999 Elsevier Science B.V. All rights reserved.

Keywords: Spitsbergen; paleomagnetism; Permian; apparent polar wandering; paleogeography

1. Introduction

The post-Permian paleotectonic position of Spitsbergen is still a matter of a debate. Rowley and Lottes [1] considered Svalbard to be a separate microplate during the Mesozoic. Malod and Mauffret [2] hypothesized paleo-north to south rifting between Spitsbergen and Norway. As remarked by Van der Voo [3], such a solution implies that pre-rifting Spitsbergen paleomagnetic poles would not be valid for Stable Europe. Devonian and Early Carboniferous poles from Spitsbergen [4,5] fit well on the Baltic apparent polar wander path (APWP), but Late Carboniferous–Permian poles [6,7] deviate from that APWP. This deviation has been explained as a result of rotation of Spitsbergen in relation to Europe during the opening of the Labrador Sea and Greenland–Norwegian Sea (op. cit.). It should be stressed, however, that in many paleotectonic maps (e.g. [8]) such post-Caledonian movement of Spitsbergen relative to the European plate has not been taken into account.

A paleomagnetic investigation of Upper Gzhelian (Upper Carboniferous) through Spathian (Lower Triassic) sediments from central Spitsbergen has been carried out in order to (1) refine the Permian–Early Triassic paleotectonic and paleogeographic position of Spitsbergen, (2) to interpret the duration and structure of the reversed-polarity Kiaman superchron. The aim of this paper is to present new paleomagnetic poles that clearly define the Permian and Early Triassic paleopositions of Spitsbergen.

2. Geological setting

Permian and Triassic rocks form a narrow belt in the western part of Spitsbergen. They were strongly

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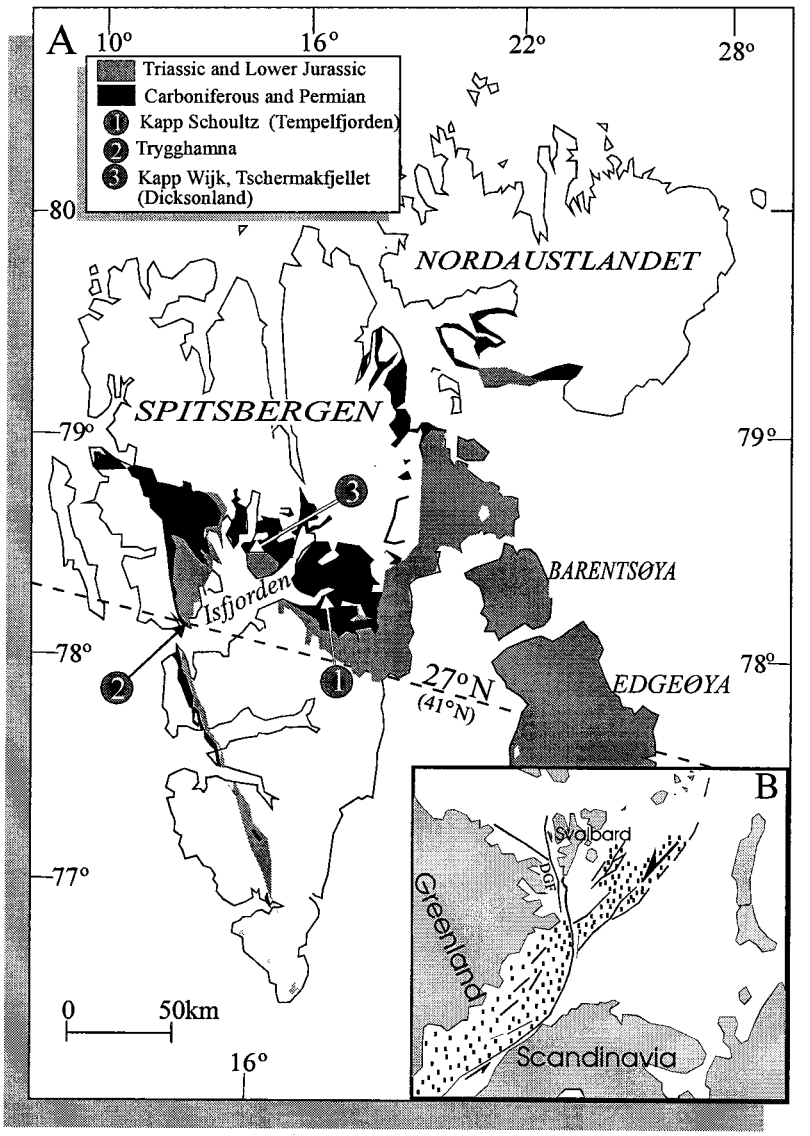


Fig. 1. (A) Carboniferous–Lower Jurasic outcrop areas in the Svalbard archipelago (after [11,12]). Regions of paleomagnetic investigations have been marked by arrows. The paleolatitude obtained from the Late Gzhelian–Kazanian samples (27ºN) is indicated by a broken line. The paleolatitude value for the Permian–Triassic boundary established from the Stable European APWP is presented in the parentheses. (B) The Barents Shelf and Svalbard during Early Cretaceous to Late Jurassic time (after [1], simplified). Domains of extension are marked by dots. *DGF*, De Geer Fault.

deformed during West Spitsbergen Orogeny [9]. Their present dips vary between 60 and 90º. Rocks of Permian and Triassic age are also exposed in the central and eastern parts of the island (Fig. 1) where their tectonic disturbance is very small (dip 0–15º). The West Spitsbergen Orogen was formed most probably during Eocene ocean spreading processes accompanied by dextral strike-slip movement along the De Geer Fault that separates Svalbard and the Barents Shelf from Greenland [8,10].

The stratigraphy of the sequence of interest to this paper is shown in Fig. 2. Moscovian–Sakmarian stages at Spitsbergen consist of limestones, dolomites and sporadic sandstones. They

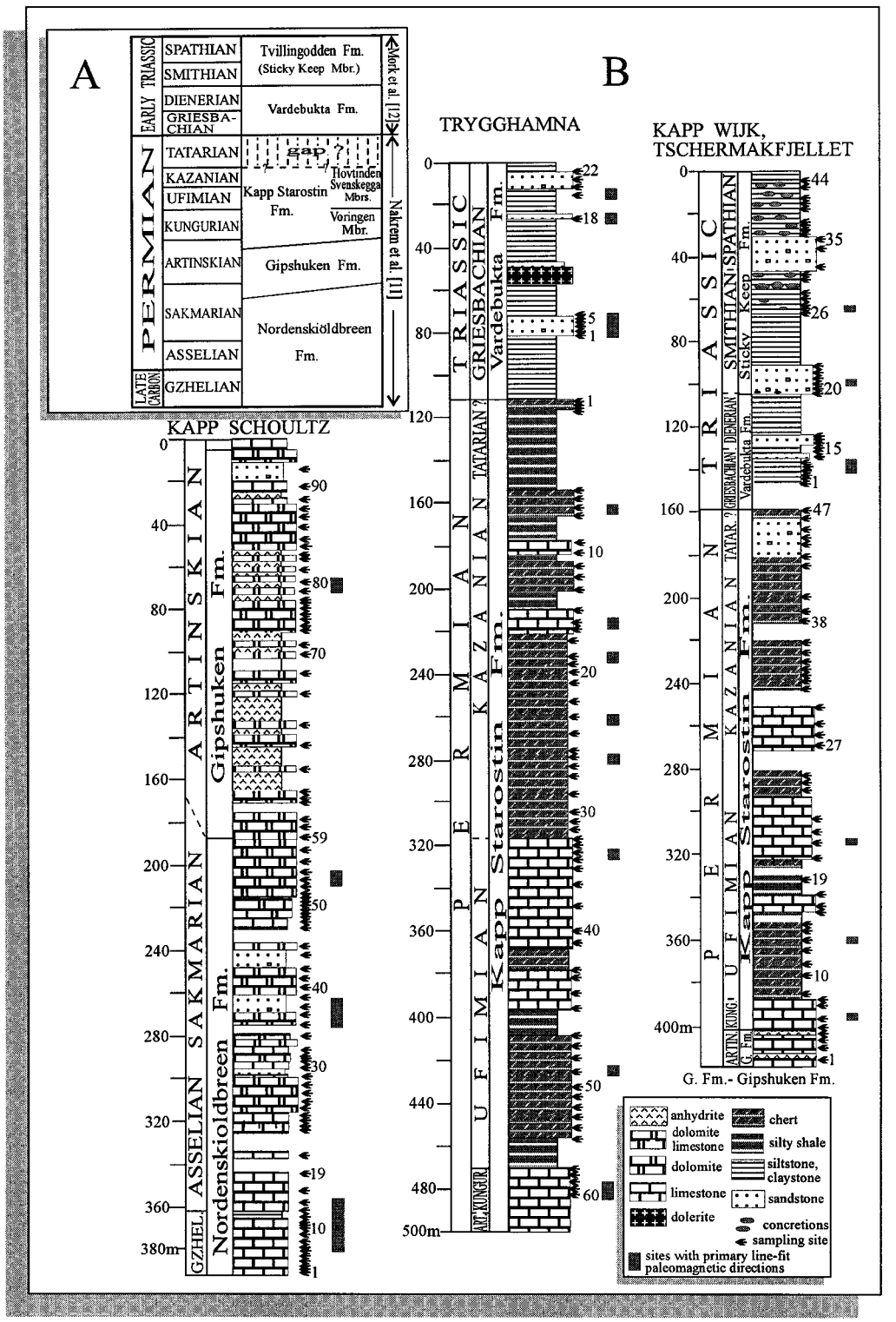


Fig. 2. (A) Late Carboniferous–Early Triassic stratigraphy in Spitsbergen. (B) Lithological columns and stratigraphy of studied sections. Arrows show sites of paleomagnetic sampling.

are classified as the Nordenskiöldbreen Formation (Fig. 2). This formation is covered by mainly evaporitic sediments of the Gipshuken Formation attributed to the Sakmarian and Artinskian [11]. These evaporitic series are overlain by cherts, spiculitic shales and limestones with rich brachiopod and bryozoan faunas, together defined as the Kapp Starostin Formation (KSF). Biostratigraphic data supports the Kungurian–Kazanian age of the KSF, whereas there is no biostratigraphic evidence to indicate the presence of Tatarian stage in the KSF [11]. It should be stressed, however, that the last 30–50 m of this formation do not contain any diagnostic fauna. The maximum thickness of the whole Permian sequence reaches about 800 m. In the Triassic period Spitsbergen was part of an extensive boreal basin. 250 to 1200 m of clastic sediments with minor intercalations of carbonate rocks were deposited during numerous transgressive–regressive cycles. The Lower Triassic succession consists of two (Fig. 2), or locally three major cycles which often consist of several subcycles [12]. Biostratigraphic documentation of these sediments [13–15], based mainly on ammonoids, together with sequence stratigraphy markers allow to subdivide them into standard chronostratigraphic units.

3. Sampling localities and paleomagnetic procedure

A total of 297 samples for paleomagnetic studies were drilled from the Isfjord area. Uppermost Carboniferous and Lower Permian rocks (Nordenskiöldbreen and Gipshuken Formations) were sampled in the Tempelfjorden (locality Kapp Schoultz; Figs. 1 and 2). Kapp Sarostin Formation and Early Triassic sediments were examined in western Dicksonland (locality Kapp Wijk and Tschermakfjellet) and near the entrance to Isfjord (locality Trygghamna). In the Tempelfjorden and Dicksonland areas the rocks studied were characterized by a very small dip that did not exceed 10º. A different tectonic situation is found in the Trygghamna area where the beds are steeply (60–85º) inclined towards NEE (KSF) or NE (Vardebukta Fm.) direction.

The natural remanent magnetization (NRM) of specimens was measured using spinner and cryogenic magnetometers. Some pilot samples were subjected to an alternating field demagnetization experiment. Because this method was not effective, the majority of the sample set has been subjected to stepwise thermal demagnetization in a μ-metal shielded oven, which reduced the ambient field close to a few nT. After each thermal demagnetization level the magnetic susceptibility signal was monitored. Least-square line fit methods, as presented by Kirschvink [16], were used to calculate the components of NRM and their unblocking temperature spectra. Thermomagnetic analysis and isothermal remanent magnetization (IRM) acquisition were used to determine the nature of magnetic carriers. For some samples, hysteresis loops were also determined.

4. Results of paleomagnetic analysis

Only a small subset of the samples yielded well-defined, antipodal directions of presumed Permian–Triassic age that were useful for paleopole calculations; however, about half of the samples yielded polarity indications. During thermal demagnetization between 380ºC and 435ºC, an abrupt increase of magnetic susceptibility was observed in the bulk of samples, and their demagnetization was terminated at this point. In many samples from all sections only one distinct component ‘A’ with very steep, positive inclination (75–90º) was isolated (Fig. 3a). This component results most probably from a widespread remagnetization that affected Spitsbergen in Cenozoic time (e.g. [17]). Some samples from a narrow tectonic zone which cuts the Trygghamna section also contain steep component ‘A’, but with a negative inclination (Fig. 3a, sample th52). In this tectonic zone, the NRM intensities were about an order of magnitude higher than those in the remagnetized samples with a positive inclination.

About 50% of the samples also revealed a component ‘B’ with moderate angles of negative or positive inclination. The degree of remagnetization of these samples varied, as demonstrated by the presence of component ‘A’. In 12% of the sample collection, component ‘B’, which is interpreted as primary magnetization vector, was well defined (Fig. 3b,c) and could be calculated by line fitting. NRM intensities of these weakly magnetized samples never exceeded

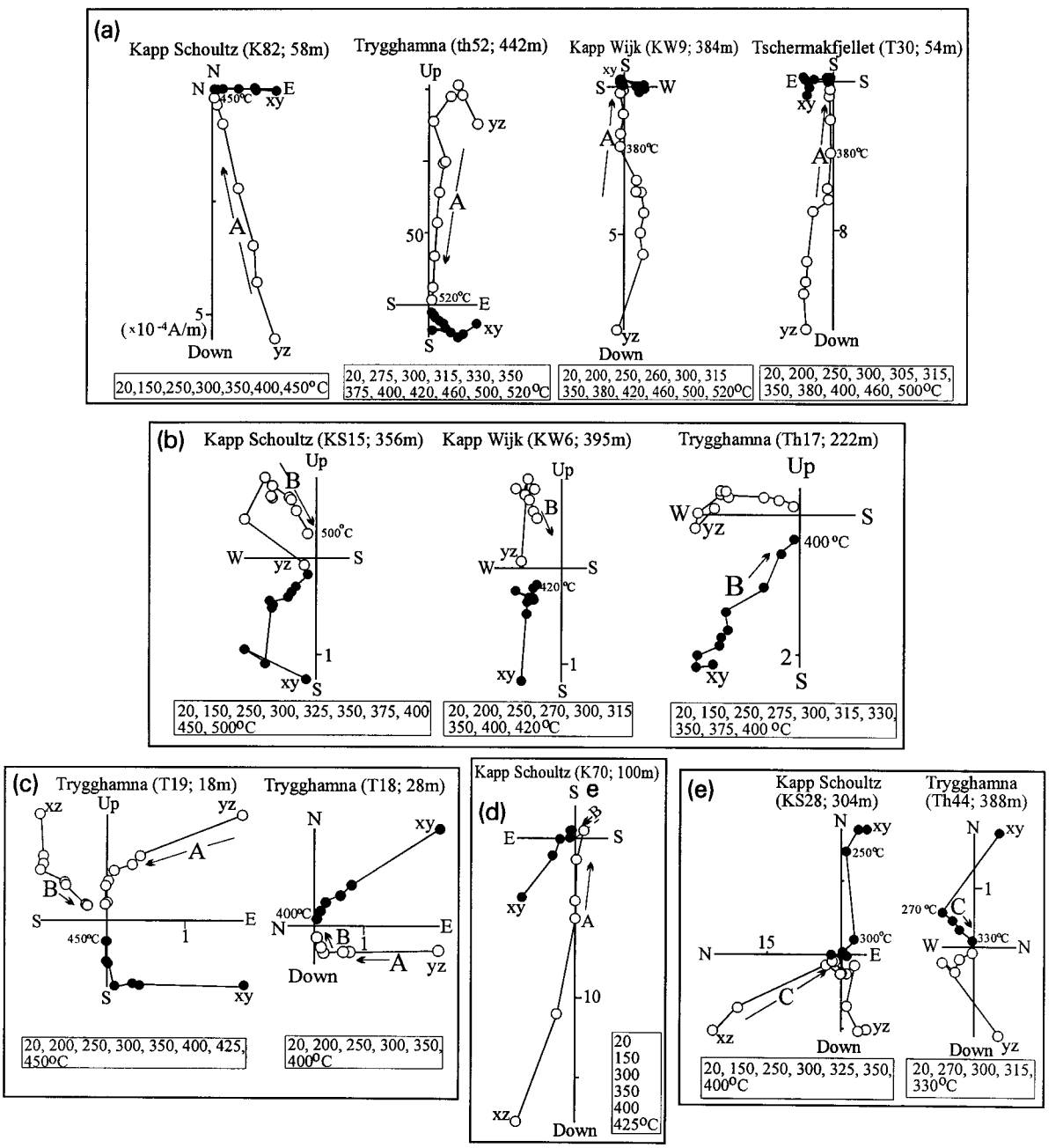


Fig. 3. Orthogonal demagnetograms of representative samples of Late Gzhelian–Early Triassic rocks from central Spitsbergen. (a) Totally remagnetized samples with Cenozoic component *A* (in situ coordinates). (b) Samples containing component *B* (after bedding correction) of Late Carboniferous–Permian age. (c) Samples with component *B* (after bedding correction) of Early Triassic age. (d) Sample containing component *B* that is strongly overlapped by component *A* (after bedding correction). (e) Samples revealing the presence of low-stable component *C* (after bedding correction).

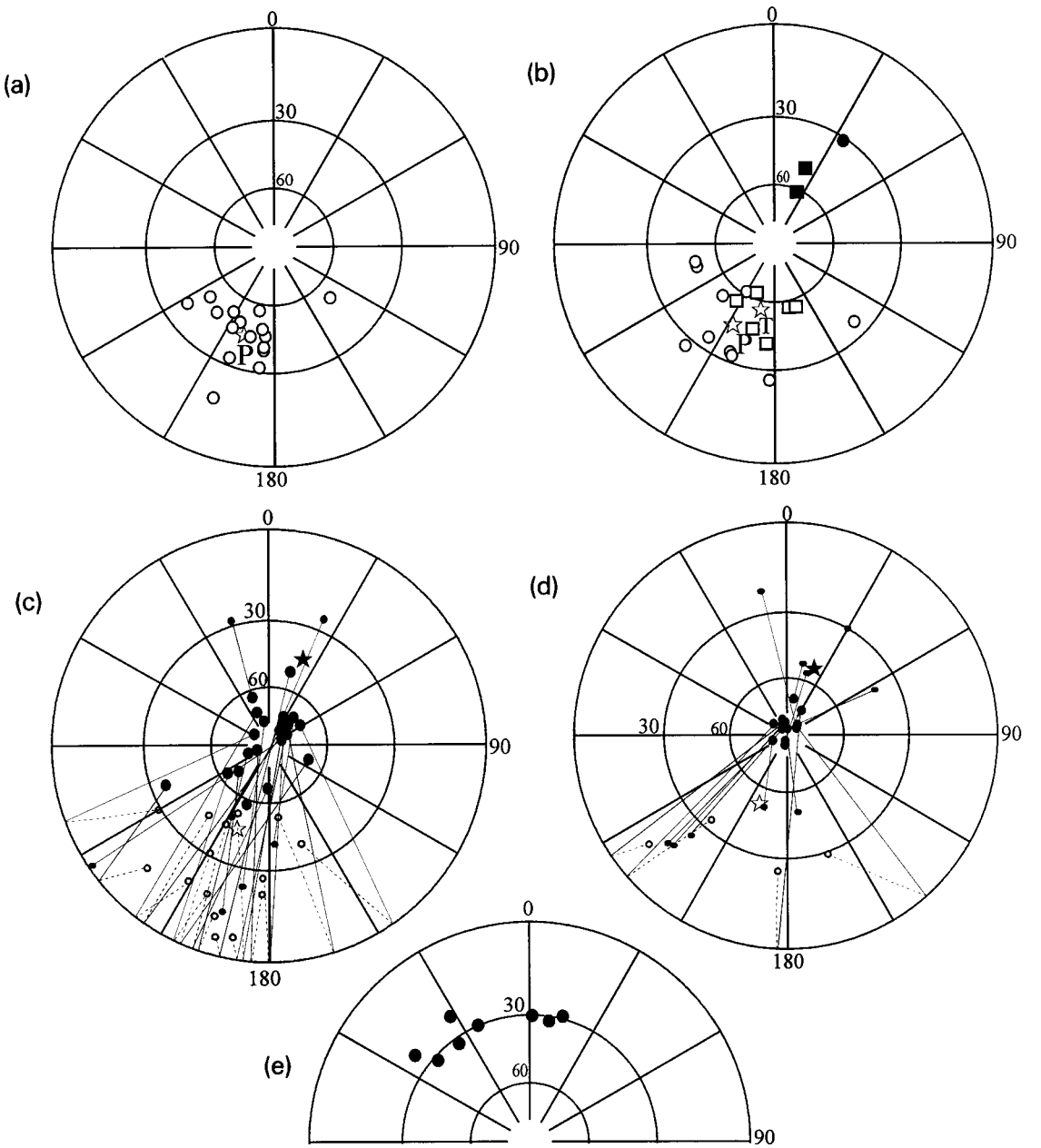


Fig. 4. (a) Stereographic projection of the Late Gzhelian–Early Permian line-fit directions from Kapp Schoultz section. The open star with letter *P* shows the mean direction. (b) Stereographic projection of Permian (circles) and early Triassic (squares) line fitted directions from Trygghamna section. Open stars (*P* and *T*) show the mean Permian and early Triassic direction respectively. (c) Stereographic projection of NRM directional tracks during thermal demagnetization of the late Gzhelian–Artinskian partly remagnetized samples from Kapp Schoultz section. Bigger circles show the NRM directions. Smaller circles present the directions obtained before rapid increase of magnetic susceptibility at temperatures of about 425°C. Open stars show expected position of normal and reversed Permian directions. (d) Stereographic projections of NRM directional tracks during thermal demagnetization of Early Triassic partly remagnetized samples from Tschermakfjellet section. (e) Stereographic projection of low temperature directions *C* isolated in Kapp Schoultz and Trygghamna sections. All directions are presented after bedding tilt correction. In stereoplots, open (closed) symbols denote upward (downward) pointing inclinations.

Table 1
Statistics of late Gzhelian–Griesbachian paleodirections isolated in the Isfjorden area

Area	Age of rocks	*N*	*D*	*I*	α_{95}	*K*	Lat.	Long.	dp	dm
Kapp Schoultz	Late Gzhelian–Artinskian	16	199	−45	8.7	19.1	38ºN	176ºE	7	11
Trygghamna	Kungurian–Kazanian	10	209	−45	17	8.6	37ºN	162ºE	14	22
			100	*−39*	*24*	*5.1*				
Tryghamna, Kapp Schoultz, Kapp Wijk (summary of stat.)	Late Gzhelian–Kazanian	29	203	−45	7.6	13.5	38ºN	171ºE	6	10
			173	*−57*	*15*	*4.1*				
Trygghamna	Griesbachian	8	191	−55	8.9	39.3	48ºN	182ºE	9	13
			92	*−20*	*16*	*13.1*				

N = number of samples; *D* = declination; *I* = inclination; α_{95}, *K* = Fisher statistics parameters [18]; Lat. = latitude of north paleomagnetic pole; Long. = longitude of north paleomagnetic pole; dp = error of the distance between site and paleopole; dm = paleodeclination error. Parameters before bedding correction are witten in *italic*.

3×10^{-4} A/m, which is at least three times lower than intensities in the samples containing only the component 'A'. The maximum unblocking temperature of component 'B' remains unknown because of the increase in magnetic susceptibility, but must be higher than 400ºC. Line-fit directions obtained in the Kapp Schoultz section cluster well in the upward, southwest quadrant. Likewise, the line-fit 'B' directions isolated in ten Permian and eight Triassic samples from Trygghamna section display a similar trend after tectonic correction (Fig. 4a,b; Table 1). Three of these 'B' directions from Trygghamna are antipodal, downward toward northeast. Prior to tectonic correction, all 'B' directions from this locality are shallow upward to the east (Table 1), in contrast to the expected postfolding Cenozoic overprint inclinations which should be almost vertical.

Most of the samples were strongly remagnetized, but lost their component 'A' at temperatures higher than 400ºC (Fig. 3d). The endpoint directions, defined before a rapid increase of magnetic susceptibility, did not attain the expected Permian–Triassic directions. Nevertheless these endpoint directions and trends of demagnetization paths indicated the polarity of the underlying component 'B' (Fig. 4c,d).

Eight samples from Tryghamna and Kapp Schoultz revealed the presence of another component of magnetization. This component, labeled 'C' and probably associated with ferric sulfides, was removed at temperatures up to 330ºC (Fig. 3e). Directions related to the component 'C' were streaked along a small circle of 30º inclination (Fig. 4e). This inclination is significantly (ca. 15º) shallower than the expected Permian inclination. For that reason it is difficult to explain the origin of component 'C', and it probably does not represent an actual geomagnetic field direction.

5. Magnetic carriers

Most IRM acquisition curves show only the presence of low coercivity magnetic minerals. IRM acquisition and thermomagnetic (orthogonal-axes) plots drawn up for partly remagnetized (sample KW47–grey chert) and totally remagnetized (sample T38 — yellow–grey siltstone) rocks from Kapp Wijk–Tschermakfjellet sections are presented in Fig. 5. The totally remagnetized Triassic sample (T38) contains a certain admixture of higher coercivity mineral, but with maximum unblocking temperature analogous to the low coercivity carrier that has been completely demagnetized between 500ºC and 550ºC. In this sample magnetite grains of a different size could carry the IRM.

Another shape of thermomagnetic curves can be observed in the case of the partly remagnetized Permian sample KW47 where the maximum unblocking temperatures do not exceed 500ºC and two earlier sharp drops of IRM intensity took place near 300ºC and 430ºC. The first drop may be related to the presence of ferric sulfides or more titanium-rich magnetite. The Permian direction in the sample KW47 remained completely uncovered until 425ºC. Hence it is very probable that Permian primary remanence resides in magnetite or titanomagnetite grains with

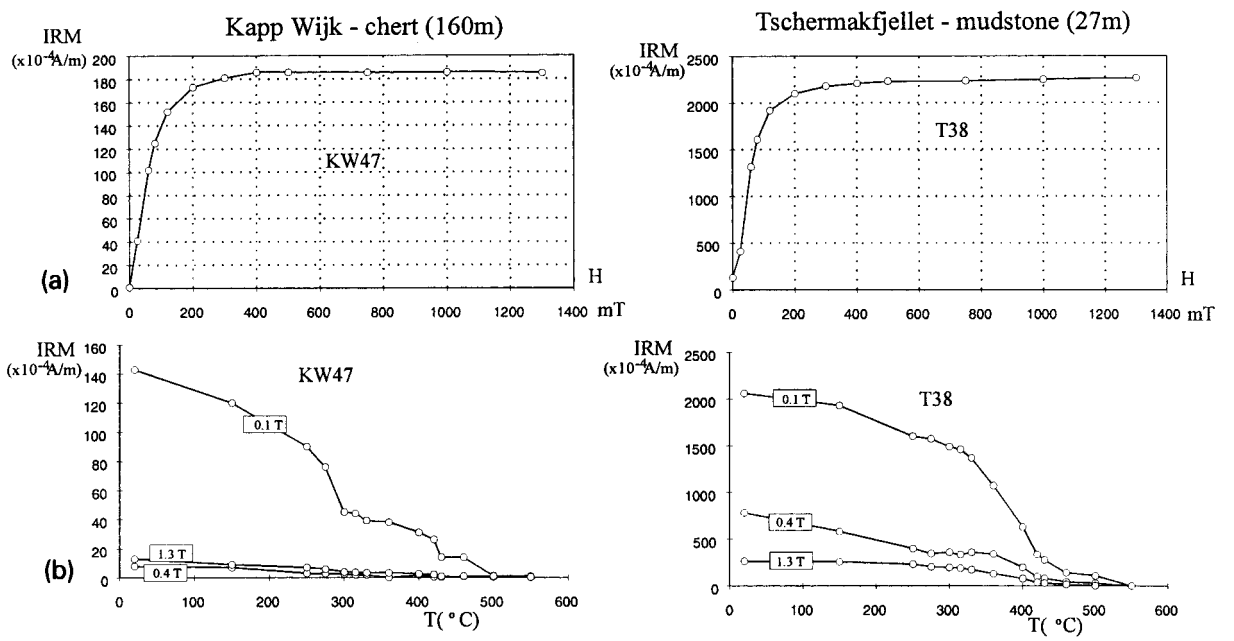


Fig. 5. (a) IRM acquisition curves, (b) thermal demagnetization of orthogonal-axes IRM curves obtained for partly (KW47) and totaly (T38) remagnetized samples from Kapp Wijk–Tschermakfjellet section.

unblocking temperatures between 450°C and 500°C. It is not easy to identify the magnetic carrier of the component 'A' that partly covered the component 'B' and was demagnetized at temperatures of about 400°C. A small decrease in magnetic susceptibility values noted in some samples at temperatures 350–400°C could suggest the presence of maghemite in the rocks studied.

A substantial increase in magnetic susceptibility at temperatures 420–440°C took place in most of the samples containing a well-defined component 'B'. This is caused most probably by the transformation of iron-bearing clay minerals to magnetite or oxidation of ferric sulfides also to magnetite [19,20]. Magnetic susceptibility did not increase during demagnetization of the totally remagnetized samples. Therefore it is likely that total remagnetization is due to oxidation of ferric sulfides to magnetite with unblocking temperatures between 500°C and 550°C.

Some carbonate samples from Kapp Schoultz and Trygghamna also contained a very high coercivity mineral (Fig. 6a). A thermomagnetic curve prepared after saturation in a field of 3 tesla (Fig. 6b) displays here the presence of goethite. This observation has been supported by the shape of hysteresis loops prepared at room temperature and after heating at 250°C. A wasp-waisted shape of the first loop points to the presence of two different coercivity magnetic minerals [21]. This form disappears after heating at 250°C (Fig. 6c,d).

Measurements of anisotropy of magnetic susceptibility (AMS) in the partly remagnetized Permian samples reveal bedding-perpendicular minimum susceptibility axes (Fig. 6e). This distribution can correspond to the primary sedimentary fabrics in natural sediments (e.g. [20]). The directions of AMS axes of totally remagnetized samples are strongly scattered (Fig. 6f).

6. Paleomagnetic poles

Only 'B' directions defined by best-fitted lines [16] were used for calculations of paleomagnetic poles. Paleopoles computed for individual localities and the mean Late Gzhelian–Kazanian paleopole are listed in Table 1. Due to a small number of Permian ($N = 3$) and Triassic ($N = 1$) fitted line directions no paleopole was calculated for Kapp Wijk–Tschermakfjellet locality.

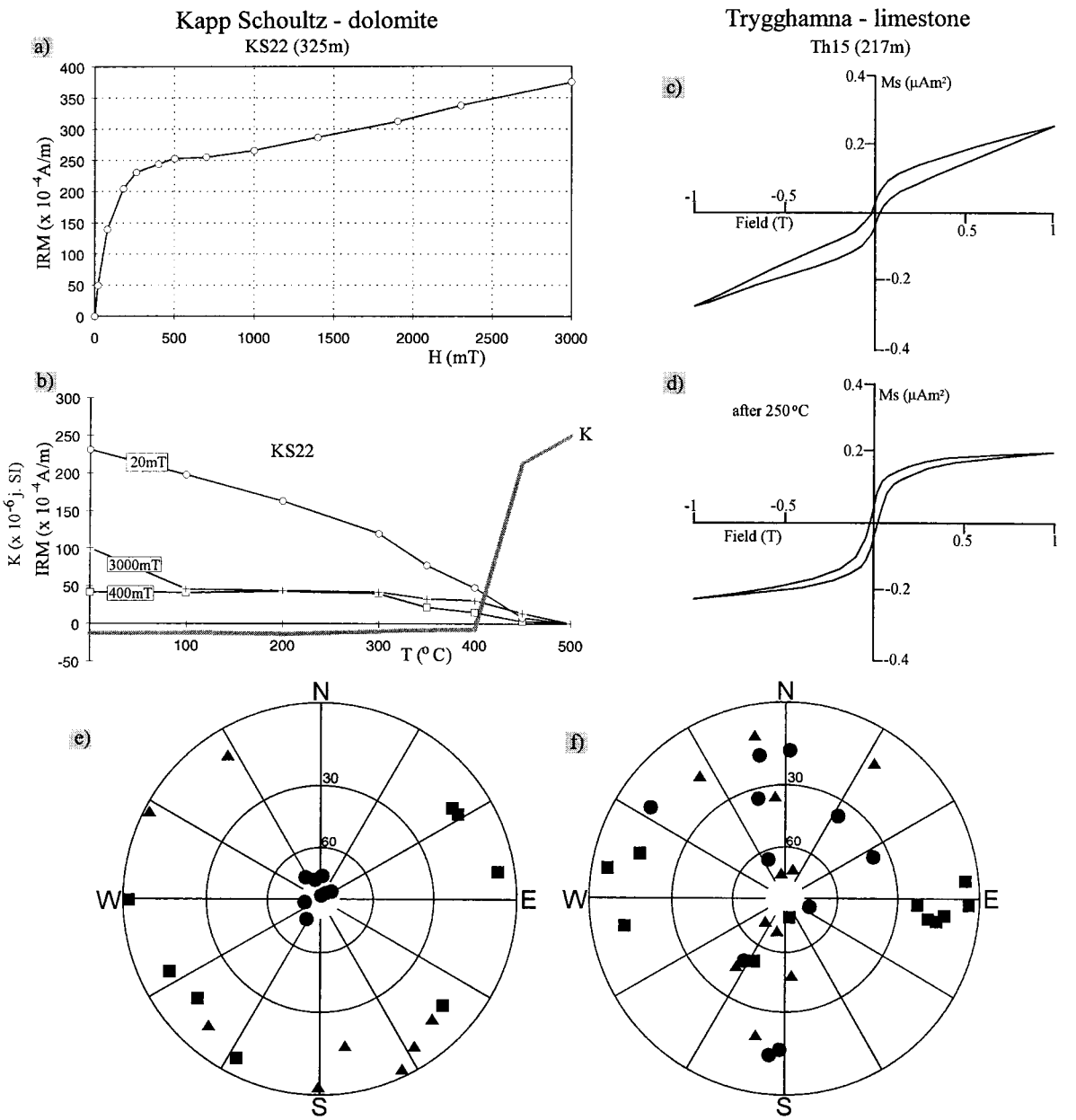


Fig. 6. (a) IRM acquisition curve, (b) thermal demagnetization of orthogonal-axes IRM and magnetic susceptibility versus demagnetizing temperature curves obtained for a dolomite sample from the Kapp Schoultz section. IRM and thermomagnetic curves show the presence of goethite. (c) Hysteresis loop for a limestone sample from the Trygghamna section. (d) Hysteresis loop for the same sample but after its heating at 250ºC. (e) Stereographic presentation of principal axes of maximum (squares), intermediate (triangles) and minimum (circles) susceptibility for partly remagnetized Permian sediments from the Kapp Schoultz and Kapp Wijk sections. (f) Stereographic projection of principal susceptibility axes for totaly remagnetized Permian sediments from the Kapp Schoultz and Kapp Wijk sections.

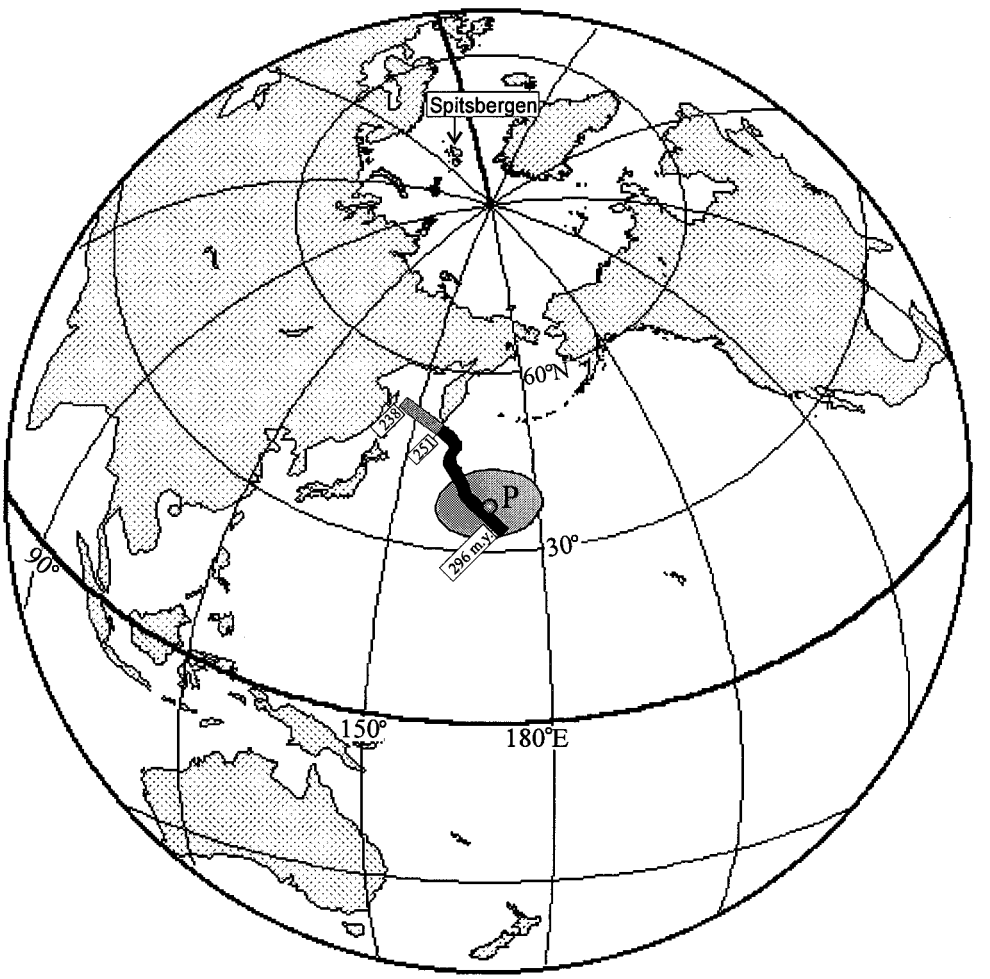


Fig. 7. Permian–Early Triassic Apparent Polar Wander Path for Baltica and the Late Gzhelian–Kazanian paleopole (labeled *P*) from this paper. The APWP was constructed using good quality poles ($Q > 3$) listed by Van der Voo [3] and the GMAP display package [25].

The mean Late Gzhelian–Kazanian paleopole has been compared with the apparent polar wander path (APWP) characteristic for Stable Europe. This APWP has been constructed on the basis of European paleopoles listed by Van der Voo [3]. Additionally, the middle Triassic paleopole of Theveniaut et al. [22] has been included. It is clearly visible that the Late Gzhelian–Kazanian paleopole from Spitsbergen fits very well to the early Permian (296 Ma) segment of the Stable European APWP (Fig. 7). The Griesbachian paleopole from Tryghamna is based only on eight samples (Table 1) and slightly departs to the east from the reference APWP (251 Ma position). This departure is due to very poor statistic representation (not sufficient number of samples) or may be related to local tectonic rotations of Triassic beds. The bedding strike of studied Griesbachian strata departs counterclockwise by about 15° from the regional bedding strike observed in the Permian strata.

The Late Gzhelian–Kazanian paleopole supports the thesis that Spitsbergen has been an integral part of the European plate at least since the Permian. Mesozoic early rifting processes between Spitsbergen and Scandinavia's mainland [2] did not take place or paleotectonic rotations due to that rifting

were below resolution of paleomagnetic method. As noted previously, the Devonian [4] and the early Carboniferous [5] paleopoles from Spitsbergen also integrate well with the Stable European APWP. It should be stressed that, in contrast to the Permian part of that APWP, the Devonian and Early Carboniferous segment of the Baltic APWP is still poorly defined. Thus the Spitsbergen Late Gzhelian–Kazanian pole provides a much tighter case for plate tectonic continuity to Europe.

The paleolatitudinal data are very important in paleoecological studies and models of mass extinctions. The paleolatitudes of Spitsbergen hitherto assumed as Kazanian (45ºN) and early Griesbachian (50ºN) were taken [23,24] from the synthetic paleogeographic maps [26]. It can now be established from the Stable European APWP that at the P–Tr boundary central Spitsbergen was located around 40ºN. The Late Gzhelian–Kazanian mean inclination (45º) obtained in this paper corresponds to 27º of paleolatitude as indicated in Fig. 1.

7. Conclusions

Paleomagnetic studies of the Permian–Early Triassic rocks from central Spitsbergen reveal widespread remagnetization. In spite of the extensive Cenozoic remagnetization, the Permian and Triassic components residing in magnetite could be resolved in several samples of the suites. The paleomagnetic pole obtained from the Late Gzhelian–Early Permian samples is in agreement with the Early Permian segment of apparent polar wander path for Baltica. It does not support the hypothesis about post-Carboniferous individual mobilism of Spitsbergen, i.e. a movement of Spitsbergen with respect to Stable Europe.

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